

AssessClear AI Pre-Pilot Master Synthesis Report

Generated directly by Qwen 3 (14B) on NVIDIA GeForce RTX 2070 SUPER GPU

MODEL ENGINE	COMPUTE ARCHITECTURE	DISTRICTS EVALUATED	STUDENTS ASSESSED
Qwen 3 (14B - Q4_K_M)	NVIDIA RTX 2070 SUPER (CUDA)	Katni, Indore, Bhopal, Raisen	~351 (Grades 6–8, Math & Hindi)

1. Executive Summary

AssessClear AI, a digital assessment tool, was piloted across 9 schools in 4 districts (Katni, Indore, Bhopal, Raisen) to evaluate its efficacy in automating answer-sheet analysis, providing personalized feedback, and supporting educators. While the tool demonstrated **strong potential for remedial learning and multi-tiered feedback**, critical technical and usability challenges emerged, including **OCR inaccuracies, dashboard limitations, network dependency, and inconsistent AI judgments**. This report synthesizes findings, empirical metrics, and recommendations for improvement.

2. Key Findings

2.1 Dashboard & Session Management Issues

- Evaluation Count Mismatch:** The dashboard counted **questions evaluated** (e.g., 29 evaluations for 3 answer sheets), not **answer sheets assessed**.
- Data Retention Limit:** A 200-evaluation cap caused older data to disappear, with **23.4% data loss** overall (59.3% in Katni).
- Session Code Retrieval:** Users struggled to retrieve session codes for subsequent assessments, requiring **manual note-taking**.
- Dashboard Usability:** Hindi subject and higher grades (7–8) were missing in Indore; history questions were inconsistent.

2.2 Technical Challenges

- OCR & Image Recognition Errors:**
 - Digit Recognition Failures:** 5.1% error rate (e.g., 9996 → 3226).
 - MCQ Misidentification:** AI misread Option D as B; unattempted questions were sometimes marked as correct.
 - Blurry Image Handling:** AI attempted to interpret unclear images instead of flagging them.
- Question Detection Failures:** 10.7% of questions skipped (e.g., Q1/Q2 on first page in Indore/Katni).
- Network Dependency:** In Raisen (MS Kharbai), **10% zero-network failure** stalled scanning and link opening.

2.3 Feedback Quality & Pedagogical Alignment

- Strengths:**
 - Multi-tiered Feedback:** *"What you did well," "Think about this," "Next steps"* were praised for remedial learning.

- **Hindi Language Capability:** Strong for MP context, though **language accuracy needed improvement**.
- **Weaknesses:**
 - **AI vs. Teacher Judgment Discrepancies:** AI marked a student's "3 months" as incorrect for a question expecting "90 days."
 - **Actionability Gap:** Feedback often focused on "**what to improve**" rather than "**how to do it**" (e.g., "practice antonyms" without classroom activity suggestions).
 - **Inconsistent Feedback Across Pages:** Teacher vs. student feedback in PDFs diverged.

2.4 Time & Resource Burden

- **Processing Time:**
 - AI processing: **38.9s/sheet** (Indore: 22.8s, Bhopal: 73.4s).
 - Human review: **144.5s/sheet** (2.4 min/paper).
 - **Total Mean Time:** 183.4s/sheet (3.06 min).
- **Class-Level Time Burden:**
 - 40-student class: **122.3 min** (2.04 hrs) without summary UI; **23.3 min** with summary UI (81% time saved).

3. Empirical Metrics (TabFM Data)

Metric	Value
Total Students Assessed	351 (Bhopal: 125, Indore: 100, Katni: 91, Raisen: 35)
AI Processing Time	38.9s/sheet (Indore: 22.8s, Bhopal: 73.4s, Katni: 26.6s)
Human Review Time	144.5s/sheet (2.4 min/paper)
Total Time per Sheet	183.4s (3.06 min)
Question Detection Fidelity	89.6% (Q1/Q2 skipped in Indore/Katni)
Digit OCR Error Rate	5.1% (e.g., 9996 → 3226)
200-Cap Data Loss Rate	23.4% overall (59.3% in Katni)
Zero-Network Failure Rate	10.0% (MS Kharbai, Raisen)
Classification Risk Triage	Critical Blocker: 10.0% (N=35), High Risk: 39.3% (N=138)

4. User Feedback & Recommendations

4.1 From Educators & Observers

- **Positive Aspects:**
 - **Remedial Learning Potential:** AI's step-by-step feedback for incorrect answers was praised.
 - **Offline Support Requested:** 100% of districts highlighted network dependency as a critical barrier.
- **Critical Concerns:**
 - **Teacher Override Needs:** AI should be a **support tool**, not a replacement (Indore, Bhopal).
 - **Competency-Based Dashboards:** Users requested drill-down by student/competency/question.
 - **Confidence Scores:** Needed to prioritize high-risk errors.

4.2 Recommendations

1. **Enhance OCR & Image Handling:**
 - Improve digit recognition (e.g., 9996 → 3226) and blur detection.
 - Flag unclear images instead of attempting interpretation.
2. **Dashboard Improvements:**
 - Fix evaluation count mismatch and data retention limits.
 - Add Hindi language accuracy checks and support for higher grades (7–8).
3. **Offline & Low-Network Support:**
 - Enable sync for 30–40+ students and open-ended questions.
 - Reduce dependency on stable internet (critical in Raisen).
4. **Teacher Override & Customization:**
 - Allow teachers to review/edit AI-generated feedback.
 - Add confidence scores for high-risk errors.

5. Time Efficiency:

- Implement class-level summary dashboards to reduce review time by 81% (from 122.3 min to 23.3 min for 40 students).

5. Conclusion

AssessClear AI shows **strong potential** for transforming assessment workflows with its **multi-tiered feedback** and **Hindi language capabilities**. However, **technical limitations** (OCR errors, dashboard flaws, network dependency) and **pedagogical alignment gaps** (feedback quality, teacher override) must be addressed to realize its full potential. Prioritizing **offline support, OCR accuracy, and teacher collaboration** will be critical for scalability and adoption in diverse educational contexts.
